

Miguel de Navascués - CV

2026-04-21

RESEARCH

Career

- 2009–present** Researcher, INRAE UMR1062 Centre de Biologie pour la Gestion des Populations (Montpellier, France).
- 2021–2023** Visiting Researcher, Human Evolution, EBC, Uppsala University (Sweden).
- 2019–2021** Marie Skłodowska-Curie Fellow, Human Evolution, EBC, Uppsala University (Sweden).
- 2007–2009** Postdoctoral researcher, CNRS UMR7625 Écologie et Évolution (Paris, France).
- 2006–2007** Visiting researcher, Université Libre de Bruxelles (Belgium).
- 2006** Postdoctoral researcher, Universidad Politécnica de Madrid & Centro de Investigación Forestal-INIA (Spain).
- 2001–2005** Doctoral candidate, University of East Anglia (UK).

Projects

1. **OPTIMISTII - Optimisation du contrôle de *Drosophila suzukii* via une synergie entre Technique de l’Insecte Stérile et Technique de l’Insecte Incompatible.** PARSADA, FranceAgriMer (2025–2029). Rol: Collaborator. PI: N. Rode.
2. **DevOCGen: Développement et applications de nouveaux outils pour la gestion et la conservation des populations naturelles à partir de données génomiques.** BiodivOc - Biodiversité Occitanie (2022–2025). Rol: Collaborator. PI: S. Boitard & R Leblois.
3. **MODEGRAD: Modélisation de l’effet de la sélection naturelle, du changement climatique et de la gestion forestière sur la distribution des génotypes dans des gradients environnementaux.** AMI CLIMAE 2021, INRAE (2022–2024). Rol: Co-coordinator. PI: I. Scotti.
4. **INTROSPEC: Genomic consequences and evolutionary causes of introgression in the late stages of speciation.** ANR AAPG 2019 (2020–2025). Rol: Collaborator. PI: P.-A. Crochet.
5. **PemilADAPT: Adaptive introgression in pearl millet.** ANR AAPG 2019 (2020–2025). Rol: Collaborator. PI: C. Berthouly-Salazar.

6. **TimeAdapt: Tracking genetic adaptation of populations using time-series genomic data.** Marie Skłodowska-Curie Individual Fellowships 2017 (2019–2021). [doi:10.5281/zenodo.3267084](https://doi.org/10.5281/zenodo.3267084). Role: Fellow.
7. **ABCSelection: Detection of loci under selection from temporal population genomic data through ABC random forest.** LabEx Agro, Cemeb & Numev Joint Call 2016 (2018–2019). Rol: PI.
8. **GenomeHarvest: Mobilizing biomathematics-bioinformatics and genomics-genetics to decipher genome organization and dynamics as pathways to crop improvement** Agropolis Fondation flagship project (2016–2019). Rol: Collaborator. PI A D'Hont & M Ruiz
9. **PlantEvol: Plant biodiversity in sub-Antarctic Islands: evolution, past and future in changing environments** Institut Paul-Emile Victor (IPEV) Project no. 1116 (2016–2017). Rol: Collaborator. PI: F Hennion→
10. **TriPTIC: *Trichogramma* pour la protection des cultures : pangénomique, traits d'histoire de vie et capacités d'établissement** ANR AAPG 2014 (2014–2018). Rol: Collaborator. PI: J-Y Rasplus
11. **Utilisation des longueurs d'haplotype et du déséquilibre de liaison pour l'inférence de l'histoire démographique populationnelle à partir de données génomique.** Appel à projets Jeunes Chercheurs et Animations Scientifiques 2013, Institut de Biologie Computationnelle (2014). Rol: Collaborator. PI: P Pudlo & R Leblois.
12. **SelfAdapt: Adaptation to climate change under selfing — experimental test and selection footprints in *Medicago truncatula*.** Meta-programme INRA - Adapting Agriculture and Forestry to Climate Change 2012 (ACCAF) INRA (2013–2016). Rol: Co-coordinator. PI: L. Gay.
13. **IBC: Institute de Biologie Computationnelle.** Investissements d'Avenir, Bioinformatique 2011 (2012–2017). Rol: Collaborator. PI: O. Gascuel.
14. **Interaction between ecological processes within community and population genetic structure.** Programme PHC Tournesol (2014–2015). Rol: Host.
15. **Apport des marqueurs de type RAD pour résoudre les relations phylogénétiques au sein de complexes d'espèces : le cas du genre *Thaumetopoea* .** Projet Innovant EFPA (2014). Rol: Co-coordinator. PI: C Kerdelhue.
16. **IGGiPop: Inférence en Génétique et Génomique des Populations.** Jeune Équipe INRA 2010 (2010–2013). Rol: Collaborator. PI: R. Vitalis.
17. **Diversidad Genética Neutral y Adaptativa de *Taxus baccata* en la Red de Parques Nacionales.** Ministerio de Medio Ambiente, Proyecto Parques Nacionales (2008–2009). Rol: Collaborator. PI: S. González-Martínez.
18. **TRANSBIODIV: Biodiversité trans-spécifique neutre et fonctionnelle : développements théoriques et quantification chez des organismes modèles.** ANR Biodiversité 2006 (2007–2009). Rol: Postdoc. PI: M.-L. Cariou.
19. **Developing Statistical Tools for Detection of Historical Demographic Expansions with Linked Microsatellites.** European Science Foundation, CONGEN Exchange Grant — 1142 (2006–2007). Rol: Grantee.

20. **Introgresión genética en las poblaciones relictas de *Pinus sylvestris* L. var *nevadensis* del Parque Nacional de Sierra Nevada.** Ministerio de Medio Ambiente, Proyecto Parques Nacionales (2005–2008). Rol: Postdoc. PI: J.J. Robledo-Arnuncio.
21. **Field Collection of Weevils (*Brachyderes* and *Rhinusa*) in Spain for Phylogeographic and Population Studies.** Richard Warn Memorial Fund Travel Award (2003). Rol: Grantee.

PRODUCTION

Publications

citation statistics

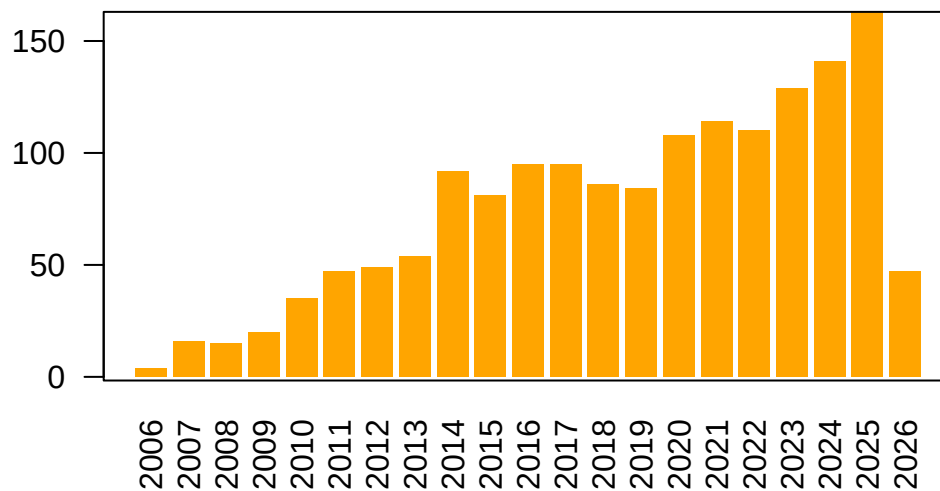


Figure 1: Number of citations per year according to Google Scholar.

preprints

articles

1. **Revisiting the African mtDNA landscape: A continental update from complete mitochondrial genomes.** I. Lankheet, A. Chowdhury, C. Tellgren-Roth, C. Jolly, A.E.R. Soares, M.d. Navascués, S. Pacchiarotti, L. Maselli, G. Kouarata, J.-P. Donzo, V. Coetzee, M.d. Castro, P. Ebbesen, E. Priehodová, E. Podgorná, V. Černý, S.T. Green, P. Harena, L. Bakrobena, F.L.M. Fomine, Z.G. Tolesa, W.A. Mengesh, M.d. Jongh, H. Soodyall, K. Bostoen, C. Barbieri, M. Larena, H. Malmström, C.M. Schlebusch (2026) [doi:10.1101/2025.04.05.647361](https://doi.org/10.1101/2025.04.05.647361). Citations: 2.
2. **Retracing the centre of origin and evolutionary history of nutmeg *Myristica fragrans*, an emblematic spice tree species.** J. Kusuma, M. Couderc, P.R. Gérard, M.d. Navascués, N. Scarcelli, J. Duminil (2026) [doi:10.1098/rspb.2025.1734](https://doi.org/10.1098/rspb.2025.1734). Citations: 6.

3. **Performance evaluation of adaptive introgression classification methods.** J. Romieu, G. Camarata, P.-A. Crochet, *M.d. Navascués*, R. Leblois, F. Rousset (2025) [doi:10.24072/pcjournal.617](https://doi.org/10.24072/pcjournal.617). Citations: 5.
4. **Analysis of the abundance of radiocarbon samples as count data.** *M.d. Navascués*, C. Burgarella, M. Jakobsson (2025) [doi:10.24072/pcjournal.522](https://doi.org/10.24072/pcjournal.522). Citations: 1.
5. **Variation in the number of radiocarbon (^{14}C) dates does not equal change in Neolithic population size over time.** A. Högborg, K. Brink, T. Brorsson, H. Malmström, E. Rudebeck, *M.d. Navascués* (2025) [doi:10.1016/j.jasrep.2025.105436](https://doi.org/10.1016/j.jasrep.2025.105436). Citations: 1.
6. **Interplay between large low-recombining regions and pseudo-overdominance in a plant genome.** M. Salson, M. Duranton, S. Huynh, C. Mariac, C. Tranchant-Dubreuil, J. Orjuela, P. Cubry, A.-C. Thuillet, C. Burgarella, *M.d. Navascués*, L. Zekraoui, M. Couderc, S. Arribat, N. Rodde, A. Barnaud, A. Faye, N. Kane, Y. Vigouroux, C. Berthouly-Salazar (2025) [doi:10.1038/s41467-025-61529-z](https://doi.org/10.1038/s41467-025-61529-z). Citations: 6.
7. **SelNeTime: a python package inferring effective population size and selection intensity from genomic time series data.** M. Uhl, P. Bunel, *M.d. Navascués*, S. Boitard, B. Servin (2025) [doi:10.1101/2024.11.06.622284](https://doi.org/10.1101/2024.11.06.622284). Citations: 2.
8. **Mating systems and recombination landscape strongly shape genetic diversity and selection in wheat relatives.** C. Burgarella, M.-F. Brémaud, G. Von Hirschheydt, V. Viader, M. Ardisson, S. Santoni, V. Ranwez, *M.d. Navascués*, J. David, S. Glémin (2024) [doi:10.1093/evlett/qrae039](https://doi.org/10.1093/evlett/qrae039). Citations: 13.
9. **Genome scan of rice landrace populations collected across time revealed climate changes' selective footprints in the genes network regulating flowering time.** N. Ahmadi, M.B. Barry, J. Frouin, *M.d. Navascués*, M.A. Toure (2023) [doi:10.1186/s12284-023-00633-4](https://doi.org/10.1186/s12284-023-00633-4). Citations: 5.
10. **Joint inference of adaptive and demographic history from temporal population genomic data.** V.A.C. Pavinato, S. De Mita, J.-M. Marin, *M.d. Navascués* (2022) [doi:10.24072/pcjournal.203](https://doi.org/10.24072/pcjournal.203). Citations: 5.
11. **Ancient and historical DNA in conservation policy.** E.L. Jensen, D. Díez-del-Molino, M.T.P. Gilbert, L.D. Bertola, F. Borges, V. Cubric-Curik, *M.d. Navascués*, P. Frandsen, M. Heuertz, C. Hvilsom, B. Jiménez-Mena, A. Miettinen, M. Moest, P. Pečnerová, I. Barnes, C. Vernesi (2022) [doi:10.1016/j.tree.2021.12.010](https://doi.org/10.1016/j.tree.2021.12.010). Citations: 71.
12. **Evolution of flowering time in a selfing annual plant: Roles of adaptation and genetic drift.** L. Gay, J. Dhinaut, M. Jullien, R. Vitalis, *M.d. Navascués*, V. Ranwez, J. Ronfort (2022) [doi:10.1002/ece3.8555](https://doi.org/10.1002/ece3.8555). Citations: 6.
13. **Power and limits of selection genome scans on temporal data from a selfing population.** *M.d. Navascués*, A. Becheler, L. Gay, J. Ronfort, K. Loridon, R. Vitalis (2021) [doi:10.24072/pcjournal.47](https://doi.org/10.24072/pcjournal.47). Citations: 7.
14. **Phylogeography of the striped field mouse, *Apodemus agrarius* (Rodentia: Muridae), throughout its distribution range in the Palaearctic region.** A. Latinne, *M.d. Navascués*, M. Pavlenko, I. Kartavtseva, R.G. Ulrich, M.-L. Tiouchichine, G. Catteau, H. Sakka, J.-P. Quéré, G. Chelomina, A. Bogdanov, M. Stanko, L. Hang, K. Neumann, H. Henttonen, J. Michaux (2020) [doi:10.1007/s42991-019-00001-0](https://doi.org/10.1007/s42991-019-00001-0). Citations: 29.

15. **Linking seascape with landscape genetics: Oceanic currents favour colonization across the Galápagos Islands by a coastal plant.** Y. Arjona, J. Fernández-López, *M.d. Navascués*, N. Alvarez, M. Nogales, P. Vargas (2020) [doi:10.1111/jbi.13967](https://doi.org/10.1111/jbi.13967). Citations: 15.
16. **Pleistocene origins of chorusing diversity in Mediterranean bush-cricket populations (*Ephippiger diurnus*).** Y. Esquer-Garrigos, R. Streiff, V. Party, S. Nidelet, *M.d. Navascués*, M.D. Greenfield (2019) [doi:10.1093/biolinnean/bly195](https://doi.org/10.1093/biolinnean/bly195). Citations: 7.
17. **Structure of multilocus genetic diversity in predominantly selfing populations.** M. Jullien, *M.d. Navascués*, J. Ronfort, K. Loridon, L. Gay (2019) [doi:10.1038/s41437-019-0182-6](https://doi.org/10.1038/s41437-019-0182-6). Citations: 34.
18. **Discontinuities in quinoa biodiversity in the dry Andes: An 18-century perspective based on allelic genotyping.** T. Winkel, M.G. Aguirre, C.M. Arizio, C.A. Aschero, M.d.P. Babot, L. Benoit, C. Burgarella, S. Costa-Tártara, M.-P. Dubois, L. Gay, S. Hocsman, M. Jullien, S.M.L. López-Campeny, M.M. Manifesto, *M.d. Navascués*, N. Oliszewski, E. Pintar, S. Zenboudji, H.D. Bertero, R. Joffre (2018) [doi:10.1371/journal.pone.0207519](https://doi.org/10.1371/journal.pone.0207519). Citations: 32.
19. **Intermediate degrees of synergistic pleiotropy drive adaptive evolution in ecological time.** L. Frachon, C. Libourel, R. Villoutreix, S. Carrère, C. Glorieux, C. Huard-Chauveau, *M.d. Navascués*, L. Gay, R. Vitalis, E. Baron, L. Amsellem, O. Bouchez, M. Vidal, V. Le Corre, D. Roby, J. Bergelson, F. Roux (2017) [doi:10.1038/s41559-017-0297-1](https://doi.org/10.1038/s41559-017-0297-1). Citations: 129.
20. **Identification of gene pools used in restoration and conservation by chloroplast microsatellite markers in Iberian pine species.** E. Hernández-Tecles, J.d.l. Heras, Z. Lorenzo, *M.d. Navascués*, R. Alia (2017) [doi:10.5424/fs/2017262-9030](https://doi.org/10.5424/fs/2017262-9030). Citations: 2.
21. **Demographic inference through approximate-Bayesian-computation skyline plots.** *M.d. Navascués*, R. Leblois, C. Burgarella (2017) [doi:10.7717/peerj.3530](https://doi.org/10.7717/peerj.3530). Citations: 19.
22. **Increased fire frequency promotes stronger spatial genetic structure and natural selection at regional and local scales in *Pinus halepensis* Mill.** K.B. Budde, S.C. González-Martínez, *M.d. Navascués*, C. Burgarella, E. Mosca, Z. Lorenzo, M. Zabal-Aguirre, G.G. Vendramin, M. Verdú, J.G. Pausas, M. Heuertz (2017) [doi:10.1093/aob/mcw286](https://doi.org/10.1093/aob/mcw286). Citations: 40.
23. **Black rat invasion of inland Sahel: insights from interviews and population genetics in south-western Niger.** K. Berthier, M. Garba, R. Leblois, *M.d. Navascués*, C. Tatard, P. Gauthier, S. Gagaré, S. Piry, C. Brouat, A. Dalecky, A. Loiseau, G. Dobigny (2016) [doi:10.1111/bij.12836](https://doi.org/10.1111/bij.12836). Citations: 30.
24. **Distinguishing migration from isolation using genes with intragenic recombination: detecting introgression in the *Drosophila simulans* species complex.** *M.d. Navascués*, D. Legrand, C. Campagne, M.-L. Cariou, F. Depaulis (2014) [doi:10.1186/1471-2148-14-89](https://doi.org/10.1186/1471-2148-14-89). Citations: 9.
25. **Genetic, morphological, and acoustic evidence reveals lack of diversification in the colonization process in an island bird.** J.C. Illera, Palmero Ana M., Laiolo Paola, Rodríguez Felipe, M.Á. C., *M.d. Navascués* (2014) [doi:10.1111/evo.12429](https://doi.org/10.1111/evo.12429). Citations: 33.
26. **The complex history of the olive tree: from Late Quaternary diversification of Mediterranean lineages to primary domestication in the northern Levant.** G. Besnard, B. Khadari, *M.d. Navascués*, M. Fernández-Mazuecos, A.E. Bakkali, N. Arrigo, D. Baali-Cherif, V.B.-B.d. Caraffa, S. Santoni, P. Vargas, V. Savolainen (2013) [doi:10.1098/rspb.2012.2833](https://doi.org/10.1098/rspb.2012.2833). Citations: 341.

27. **Recent population decline and selection shape diversity of taxol-related genes.** C. Burgarella, *M.d. Navascués*, M. Zabal-Aguirre, E. Berganzo, M. Riba, M. Mayol, G.G. Vendramin, S.C. González-Martínez (2012) [doi:10.1111/j.1365-294X.2012.05532.x](https://doi.org/10.1111/j.1365-294X.2012.05532.x). Citations: 35.
28. **Evolution of neutral and flowering genes along pearl millet (*Pennisetum glaucum*) domestication.** G. Lakis, *M.d. Navascués*, S. Rekima, M. Simon, M.-S. Remigereau, M. Leveugle, N. Takvorian, F. Lamy, F. Depaulis, T. Robert (2012) [doi:10.1371/journal.pone.0036642](https://doi.org/10.1371/journal.pone.0036642). Citations: 17.
29. **Mutation rate estimates for 110 Y-chromosome STRs combining population and father-son pair data.** C. Burgarella, *M.d. Navascués* (2011) [doi:10.1038/ejhg.2010.154](https://doi.org/10.1038/ejhg.2010.154). Citations: 118.
30. **Genetic diversity and fitness in small populations of partially asexual, self-incompatible plants.** *M.d. Navascués*, S. Stoeckel, S. Mariette (2010) [doi:10.1038/hdy.2009.159](https://doi.org/10.1038/hdy.2009.159). Citations: 46.
31. **CpDNA-based species identification and phylogeography: application to African tropical tree species.** J. Duminil, M. Heuertz, J.-L. Doucet, N. Bourland, C. Cruaud, F. Gavory, C. Doumenge, *M.d. Navascués*, O.J. Hardy (2010) [doi:10.1111/j.1365-294X.2010.04917.x](https://doi.org/10.1111/j.1365-294X.2010.04917.x). Citations: 63.
32. **Combining contemporary and ancient DNA in population genetic and phylogeographical studies.** *M.d. Navascués*, F. Depaulis, B.C. Emerson (2010) [doi:10.1111/j.1755-0998.2010.02895.x](https://doi.org/10.1111/j.1755-0998.2010.02895.x). Citations: 62.
33. **Estimating gametic introgression rates in a risk assessment context: a case study with Scots pine relicts.** J.J. Robledo-Arnuncio, *M.d. Navascués*, S.C. González-Martínez, L. Gil (2009) [doi:10.1038/hdy.2009.78](https://doi.org/10.1038/hdy.2009.78). Citations: 30.
34. **Characterization of historical demographic expansions from pairwise comparisons of haplotypes using linked microsatellites.** *M.d. Navascués*, O. Hardy, C. Burgarella (2009) [doi:10.1534/genetics.108.098194](https://doi.org/10.1534/genetics.108.098194). Citations: 15.
35. **Elevated substitution rate estimates from ancient DNA: model violation and bias of Bayesian methods.** *M.d. Navascués*, B.C. Emerson (2009) [doi:10.1111/j.1365-294X.2009.04333.x](https://doi.org/10.1111/j.1365-294X.2009.04333.x). Citations: 80.
36. **The effect of altitude on the pattern of gene flow in the endemic Canary Island pine, *Pinus canariensis*.** *M.d. Navascués*, G.G. Vendramin, B.C. Emerson (2008) [doi:10.1515/sg-2008-0052](https://doi.org/10.1515/sg-2008-0052). Citations: 18.
37. **Narrow genetic base in forest restoration with holm oak (*Quercus ilex* L.) in Sicily.** C. Burgarella, *M.d. Navascués*, Á. Soto, Á. Lora, S. Fici (2007) [doi:10.1051/forest:2007055](https://doi.org/10.1051/forest:2007055). Citations: 38.
38. **Natural recovery of genetic diversity by gene flow in reforested areas of the endemic Canary Island pine, *Pinus canariensis*.** *M.d. Navascués*, B.C. Emerson (2007) [doi:10.1016/j.foreco.2007.04.009](https://doi.org/10.1016/j.foreco.2007.04.009). Citations: 42.
39. **Chloroplast microsatellites reveal colonization and metapopulation dynamics in the Canary Island pine.** *M.d. Navascués*, Z. Vaxevanidou, S.C. González-Martínez, J. Climent, L. Gil, B.C. Emerson (2006) [doi:10.1111/j.1365-294X.2006.02960.x](https://doi.org/10.1111/j.1365-294X.2006.02960.x). Citations: 75.

40. **Chloroplast microsatellites: measures of genetic diversity and the effect of homoplasy.** *M.d. Navascués*, B.C. Emerson (2005) doi:10.1111/j.1365-294X.2005.02504.x. Citations: 102.

book chapter

1. **Introgresión genética en las poblaciones relictas de *Pinus sylvestris* L. var. *nevadensis* del Parque Nacional de Sierra Nevada.** J.J. Robledo-Arnuncio, *M.d. Navascués*, S.C. González-Martínez, L. Gil (2009) in L. Ramírez, B. Asensio *Proyectos de Investigación en Parques Nacionales: 2005-2008*, Organismo Autónomo Parques Nacionales, Madrid http://www.marm.es/es/ministerio/organizacion/organismos-publicos/organismo-autonomo-parques-nacionales-oapn/prog-inv-pn/oapn_inv_artic05.aspx.

theses

1. **Modélisation intégrée des paramètres démographiques et génétiques.** T. Lejeune supervised by *M.d. Navascués*, R. Leblois, L. Marescot (2025) MSc thesis. Université de Montpellier, Montpellier (France).
2. **Comment identifier la part de l'introggression qui est due à la sélection ?** J. Romieu supervised by F. Rousset, R. Leblois, *M.d. Navascués* (2025) PhD thesis. Université de Montpellier, Montpellier (France).
3. **Exploration de scénarios évolutifs de colonisation des Îles Kerguelen par la truite commune (*Salmo trutta*).** L. Vallée supervised by S. Muratorio, *M.d. Navascués*, G. Brahy (2024) MSc thesis. Université de Pau et des Pays de l'Adour, Pau (France).
4. **Évaluation des performances des méthodes de classification d'introggression adaptative.** G. Camarata supervised by J. Romieu, F. Rousset, R. Leblois, *M.d. Navascués* (2024) MSc thesis. Université de Montpellier, Montpellier (France).
5. **Analyse temporelle de la diversité en régime autogame : Approches théorique et empirique.** M. Jullien supervised by J. Ronfort, L. Gay, *M.d. Navascués* (2019) PhD thesis. Montpellier SupAgro, Montpellier (France).
6. **Detection of archaic introgression without genomic reference from the source population an ABC approach.** E.-E. Giaglara supervised by *M.d. Navascués*, M. Gautier (2019) MSc thesis. Université de Montpellier, Montpellier (France).
7. **Évaluation de la performance des statistiques pour la détection de l'introggression adaptative chez le mil (*Pennisetum glaucum*).** O. Belkassé supervised by *M.d. Navascués*, C. Burgarella (2018) MSc project. Université de Montpellier, Montpellier (France).
8. **Limites des genome-scans pour la détection de locus sous sélection chez les espèces autogames : une étude par simulation.** A. Becheler supervised by *M.d. Navascués*, R. Vitalis (2014) MSc thesis. Université Montpellier 2, Montpellier (France).
9. **Diversification of the olive tree (*Olea europaea*) in the Central Sahara and Macaronesia during the Quaternary.** M. Goudet supervised by G. Besnard, *M.d. Navascués* (2014) MSc thesis. Université Toulouse III Paul Sabatier, Toulouse (France).

10. **Genetic diversity of the endemic Canary Island pine tree, *Pinus canariensis*.** *M.d. Navascués* supervised by B.C. Emerson (2005) PhD thesis. University of East Anglia, Norwich (UK) [doi:10.6084/m9.figshare.1108027.v1](https://doi.org/10.6084/m9.figshare.1108027.v1).
11. **Metapopulation dynamics as an explanation to the genetic structure of the desert locust populations: parameter exploration.** *M.d. Navascués* supervised by K. Ibrahim (2001) BSc project. University of East Anglia, Norwich (UK) [doi:10.6084/m9.figshare.1108026.v1](https://doi.org/10.6084/m9.figshare.1108026.v1).

conference proceedings

1. **Diversidad y flujo genético en plantaciones de *Pinus canariensis*.** *M.d. Navascués*, B.C. Emerson (2006) in *IV Jornadas Forestales de la Macaronesia*, Gobierno de Canarias, Governo dos Açores, Arena, Breña Baja, La Palma (Spain) <https://hal.inrae.fr/hal-02755419>.
2. **Estudio de la variabilidad genética de repoblaciones de *Quercus ilex* L. subsp. *ballota* (Desf.) Samp. in Bol. en Andalucía.** C. Burgarella, *M.d. Navascués*, S. Fici, Á.L. González (2005) in *Libro de Resúmenes, Conferencias y Ponencias del IV Congreso Forestal Español*, Sociedad Española de Ciencias Forestales, Zaragoza (Spain) <https://hal.inrae.fr/hal-02828959>.
3. **El éxito evolutivo de los insectos.** *M.d. Navascués* (1997) in Á. Delgado Buscalioni, J. Díaz Díaz, M. Herráiz Moreno (eds.) *I Encuentro de Estudiantes sobre Biología Evolutiva*, Universidad Autónoma de Madrid, Asociación de Estudiantes de Biología, Madrid (Spain).

Software and Data

Software

1. **SelNeTime: a python package inferring effective population size and selection intensity from genomic time series data.** M. Uhl, P. Bunel, *M.d. Navascués*, S. Boitard, B. Servin (2025) <https://forge.inrae.fr/genetic-time-series/selnetime>.
2. **DARth ABC: analysis of the dated archaeological record through approximate Bayesian computation.** *M.d. Navascués* (2024) [doi:10.5281/zenodo.13383349](https://doi.org/10.5281/zenodo.13383349).
3. **TimeAdapt: joint inference of demography and selection from longitudinal population genomic data.** *M.d. Navascués* (2023) [doi:10.5281/zenodo.8232572](https://doi.org/10.5281/zenodo.8232572).
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10. **microsatABC-IM: Approximate Bayesian computation inferences under the isolation with migration model from microsatellite data.** *M.d. Navascués* (2017) [doi:10.5281/zenodo.895905](https://doi.org/10.5281/zenodo.895905).
11. **TempoDiff: a computer program to detect selection from temporal genetic differentiation.** *R. Vitalis, L. Gay, M.d. Navascués* (2017) [doi:10.5281/zenodo.375600](https://doi.org/10.5281/zenodo.375600).
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Data

1. **VCF files from the study of low recombining genomic regions in pearl millet.** *M. Salson, M. Duranton, S. Huynh, C. Mariac, C. Tranchant-Dubreuil, J. Orjuela, P. Cubry, A.-C. Thuillet, C. Burgarella, M.d. Navascués, L. Zekraoui, M. Couderc, S. Arribat, N. Rodde, A. Barnaud, A. Faye, N. Kane, Y. Vigouroux, C. Berthouly-Salazar* (2025) [doi:10.23708/SN2T4A](https://doi.org/10.23708/SN2T4A).
2. **Medicago truncatula temporal data from: Structure of multilocus genetic diversity in predominantly selfing populations.** *M. Jullien, M.d. Navascués, J. Ronfort, K. Loidon, L. Gay* (2019) [doi:10.15454/VCZIMR](https://doi.org/10.15454/VCZIMR).
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7. **Data from: Genetic diversity of the endemic Canary Island pine tree, *Pinus canariensis* (Ph.D. thesis).** *M.d. Navascués*, B.C. Emerson (2012) [doi:10.5061/dryad.7tr51](https://doi.org/10.5061/dryad.7tr51).
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Presentations

oral communications

1. **Detection of introgression: how hard can it be?** *M.d. Navascués* (2024) in *From hybridisation to introgression: detection and evolutionary consequences*, UMR DIADE, IRD, Online, Montpellier (France). **Invited**.
2. **Demographic inference from radiocarbon data.** *M.d. Navascués* (2024) in *Mathematical and Computational Evolutionary Biology*, MCEB, Saint-Martin-de-Londres (France).
3. **Joint inference of demography and selection from genomic temporal data using Approximate Bayesian Computation.** *M.d. Navascués*, V.A.C. Pavinato, S. De Mita, J.-M. Marin (2019) in *Mathematical and Computational Evolutionary Biology*, MCEB, Porquerolles (France) <https://hal.inrae.fr/hal-02787319>.
4. **Joint inference of adaptive and demographic history from temporal population genomic data.** *M.d. Navascués*, V.A. Pavinato, S. De Mita, J.-M. Marin (2019) in *Approche Interdisciplinaire de l'Evolution Moléculaire*, GDR AIEM, Toulouse (France) <https://hal.inrae.fr/hal-02790079>.
5. **Tracking adaptation of selfing populations to environmental changes with temporal genetic data.** *M.d. Navascués* (2017) in *4èmes Rencontres Scientifiques du Département EFPA (INRA)*, Département EFPA (INRA), Pont-à-Mousson (France).
6. **Tracking adaptation to environmental changes: selfing and detection of loci under selection.** *M.d. Navascués*, A. Becheler, R. Vitalis (2015) in *Empirisme et Théorie en Écologie et Évolution*, GDR AIEM, Gif-sur-Yvette (France) <https://hal.inrae.fr/hal-02792415>.
7. **Limites des genome-scans pour la détection de locus sous sélection chez les espèces autogames.** *M.d. Navascués*, A. Becheler, R. Vitalis (2014) in *Journées ARCAD*, ARCAD, Montpellier (France) <https://hal.inrae.fr/hal-02794123>.
8. **A new composite likelihood method to characterize demographic expansions: preliminary results from Y-chromosome STR data.** *M.d. Navascués*, C. Burgarella (2008) in *DNA in Forensics*, Università Politecnica delle Marche, Ancona (Italy) <https://hal.inrae.fr/hal-02755469>.
9. **Distinguishing shared ancestral polymorphism from recent introgression in genes with recombination.** *M.d. Navascués*, F. Depaulis (2008) in *Réunion du GDR Génomique des Populations et Génomique Évolutive*, GDR Génomique des Populations et Génomique Évolutive, Sète (France) <https://hal.inrae.fr/hal-02814597>.
10. **The role of clonality in the maintenance of self-incompatibility.** *M.d. Navascués*, S. Mariette (2007) in *2èmes Journées EvoSud*, Réseau EvoSud, Orsay (France) <https://hal.inrae.fr/hal-02814599>.

11. **Characterization of demographic expansions from linked microsatellite data.** *M.d. Navascués* (2007) in *Workshop: Estimating Demographic Parameters from Genetic Data*, Conférence Universitaire de Suisse Occidentale (CUSO), La Fouly (Switzerland) <https://hal.inrae.fr/hal-02814600>.
12. **High substitution rate estimates from temporally sampled sequences: are they biased or biologically meaningful?** *M.d. Navascués*, B.C. Emerson (2007) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMOBE, Halifax (Canada) <https://hal.inrae.fr/hal-02751473>.
13. **Coalescent simulations for the evaluation of statistical tools for demographic inference and the effect of homoplasy in chloroplast microsatellites.** *M.d. Navascués*, B.C. Emerson (2006) in *Population Genetics and Genomics of Forest Trees: from Gene Function to Evolutionary Dynamics and Conservation*, IUFRO, Alcalá de Henares (Spain) <https://hal.inrae.fr/hal-02751238>.
14. **Age of alleles: how ancestral are aDNA sequences?** *M.d. Navascués* (2006) in *New Tools for Biodiversity Conservation Through the Advancement of Phylogeographic Methodologies*, European Science Foundation Exploratory Workshop, Norwich (UK) <https://hal.inrae.fr/hal-02814083>. Invited.
15. **Estudio biogeográfico y demográfico del pino canario mediante marcadores moleculares.** *M.d. Navascués*, B.C. Emerson (2005) in *XIII Workshop Avances en Biología Molecular por Jóvenes Investigadores en el Extranjero*, Centro Nacional de Biotecnología, Madrid (Spain) <https://hal.inrae.fr/hal-02827237>.
16. **Genetic diversity in the Canary Island pine tree.** *M.d. Navascués*, B.C. Emerson (2004) in *Bio-Colloquium*, School of Biological Sciences, University of East Anglia, Norwich (UK) <https://hal.inrae.fr/hal-02827238>.
17. **Can flowering time produce isolation by altitude in the Canary Island pine?** *M.d. Navascués*, B.C. Emerson (2004) in *IXth IOPB Meeting - Plant Evolution in Mediterranean Climate Zones*, IOPB, Valencia (Spain) <https://hal.inrae.fr/hal-02763094>.
18. **El éxito evolutivo de los insectos.** *M.d. Navascués* (1997) in *I Encuentro de Estudiantes sobre Biología Evolutiva*, Universidad Autónoma de Madrid, Asociación de Estudiantes de Biología, Madrid (Spain).

co-authorship on oral communications

* presenting author

1. **Joint inference of selection and demography from genomic time series.** P. Bunel* , M. Uhl, *M.d. Navascués*, S. Boitard, B. Servin (2026) in *PopGroup 59 - ALPHY*, The Genetics Society and CNRS, Lille (France).
2. **Inferring demography and selection from genomic time series data.** S. Boitard* , P. Bunel, M. Uhl, *M.d. Navascués*, B. Servin (2025) in *Premier Webinaire TIME-ID*, TIME-ID Consortium (INRAE), Online <https://hal.inrae.fr/hal-05107536>.
3. **How to identify the part of introgression that is adaptive?** J. Romieu* , M. Uhl, *M.d. Navascués*, R. Leblois, F. Rousset (2025) in *Annual Meeting of GDR Approche Interdisciplinaire de l'Évolution Moléculaire*, GDR AIEM, Saint-Martin-de-Londres (France).

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5. **SelNeTime : a new method inferring demography and selection from genomic time series data.** S. Boitard* , M. Uhl, M.d. Navascués, B. Servin (2024) in *Mathematical and Computational Evolutionary Biology*, MCEB, Saint-Martin-de-Londres (France) <https://hal.inrae.fr/hal-04659093>.
6. **Overdominance likely maintained large structural variants polymorphism in pearl millet.** M. Salson* , M. Duranton, S. Huynh, C. Mariac, C. Tranchant-Dubreuil, J. Orjuela, A.-C. Thuillet, P. Cubry, C. Burgarella, M.d. Navascués, L. Zekraoui, M. Couderc, S. Arribat, N. Rodde, A. Barnaud, A. Faye, N. Kane, Y. Vigouroux, C. Berthouly-Salazar (2024) in *Environmental and Agronomical Genomics*, France Génomique, GDR Génomique Environmental, Toulouse (France).
7. **Joint inference of demography and selection from genomic temporal data using Approximate Bayesian Computation.** V.A. Pavinato* , S. De Mita, J.-M. Marin, M.d. Navascués (2019) in *ALPHY: Bioinformatics and Evolutionary Genomics*, GDR BIM Bioinformatique moléculaire, Paris (France) <https://hal.inrae.fr/hal-02788596>.
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9. **Tracking selection in time-series population genomic data using ABC random forests.** V.A.C. Pavinato* , M.d. Navascués (2018) in *Journée scientifique Inter LabEx - « Adaptation des organismes à l'environnement »*, LabEx CeMeb, LabEx Tulip, LabEx Agro, Montpellier (France) <https://hal.inrae.fr/hal-02785695>.
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12. **Tracking selection in time-series population genomic data using ABC random forests.** V.A.C. Pavinato* , J.-M. Marin, M.d. Navascués (2018) in *II Joint Congress on Evolutionary Biology*, European Society for Evolutionary Biology, American Society of Naturalists, Society for the Study of Evolution, Society of Systematic Biologists, Montpellier (France) <https://hal.inrae.fr/hal-02785503>.
13. **Temporal study of diversity in selfing plants.** M. Jullien* , M.d. Navascués, J. Ronfort, L. Gay (2017) in *Le Petit Pois Dérivé 38*, Groupe d'Étude de Biologie et Génétique des Populations Date, Orsay (France) <https://hal.inrae.fr/hal-02788128>.
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15. **Ecological genetics and adaptation in European yew (*Taxus baccata* L.).** S.C. González-Martínez* , M. Riba, C. Burgarella, *M.d. Navascués*, F. Bagnoli, S. Cavers, D. Grivet, G.G. Vendramin, M. Mayol (2016) in *Genomics and Forest Tree Genetics Conference*, IUFRO, Arcachon (France) <https://hal.inrae.fr/hal-02741608>.
16. **Limites des genome-scans pour la détection de locus sous sélection chez les espèces autogames.** A. Becheler* , *M.d. Navascués*, R. Vitalis (2014) in *Le Petit Pois Dérivé*, Groupe d'Étude de Biologie et Génétique des Populations, Orsay (France).
17. **Population genetics of Spectacled Warbler, *Sylvia conspicillata*.** J.C. Illera* , A.M. Palmero, P. Laiolo, *M.d. Navascués* (2013) in *9th Conference of the European Ornithologists's Union*, European Ornithologists's Union, Norwich (UK) <https://hal.inrae.fr/hal-02747155>.
18. **Genetic diversity and fitness in gametophytic self-incompatible and asexual populations : theoretical approach and experimental observations in *Prunus avium* L..** S. Mariette* , *M.d. Navascués* (2010) in *Green Plant Breeding Technologies - International conference*, Vienna International Plant Conference Association, Vienna (Austria) <https://hal.inrae.fr/hal-02811680>.
19. **Evolution of neutral and flowering genes under domestication in pearl millet (*Pennisetum glaucum*).** G. Lakis* , S. Rekima, *M.d. Navascués*, F. Lamy, M.-S. Remigereau, M. Leveugle, F. Depaulis, T. Robert (2010) in *Plant and Animal Genome XVIII Conference*, Scherago International, San Diego (USA).
20. **Interaction between gene flow and selection in hybridizing oak species.** P. Garnier-Géré* , P. Abadie, *M.d. Navascués*, T. Lang, O. Brendel, C. Scotti-Saintagne, S. Gerber, C. Bodénès, R.J. Petit, A. Kremer (2009) in *Genomics of Forest and Ecosystem Health in the Fagaceae*, EVOLTREE, Research Triangle Park, North Carolina (USA).
21. **Discrimination between ancestral polymorphism and recent introgression as causes of shared polymorphic sites between species.** F. Depaulis* , *M.d. Navascués* (2009) in *International Conference on Polyploidy, Hybridization and Biodiversity*, Polyploidy Consortium 'Polyploidy and Biodiversity', Saint Malo (France).
22. **Ancient DNA and the time dependency of mutation rates.** B.C. Emerson* , *M.d. Navascués* (2007) in *41st Population Genetics Group Conference*, PopGroup, Warwick (UK).
23. **Introgresión genética en la población relictada de *Pinus sylvestris* L. var. *nevadensis* del Parque Nacional de Sierra Nevada: Implicaciones para la conservación de sus recursos genéticos.** L. Gil* , J.J. Robledo-Arnuncio, *M.d. Navascués*, S.C. González-Martínez (2006) in *Jornada de Seguimiento de Proyectos de Investigación*, Parque Nacional de Sierra Nevada, Granada (Spain).
24. **Sistemas de reproducción de las especies forestales y conservación in situ.** S.C. González-Martínez* , J.J. Robledo-Arnuncio, *M.d. Navascués* (2006) in *Simposio sobre Conservación y Uso Sostenible de Recursos Genéticos Forestales*, NA, La Laguna (Spain).
25. **Estudio de la variabilidad genética de repoblaciones de *Quercus ilex* L. subsp. *ballota* (Desf.) Samp. in Bol. en Andalucía.** C. Burgarella* , *M.d. Navascués*, S. Fici, Á.L. González (2005) in *IV Congreso Forestal Español*, Sociedad Española de Ciencias Forestales, Zaragoza (Spain) <https://hal.inrae.fr/hal-02828959>.

posters

1. **Interplay between low-recombining regions and overdominance in a plant genome?** M. Salson, M. Duranton, S. Huynh, C. Mariac, C. Tranchant-Dubreuil, J. Orjuela, P. Cubry, A.-C. Thuillet, C. Burgarella, *M.d. Navascués*, L. Zekraoui, M. Couderc, S. Arribat, N. Rodde, A. Barnaud, A. Faye, N. Kane, Y. Vigouroux, C. Berthouly-Salazar (2025) in *Congress of the European Society for Evolutionary Biology*, ESEB, Barcelona (Spain) <https://hal.science/hal-05227656>.
2. **Infer selection and demography from genomic time series.** P. Bunel, M. Uhl, *M.d. Navascués*, S. Boitard, B. Servin (2025) in *Séminaire des Doctorants du Département de Génétique Animale de l'INRAE*, Département GA (INRAE), Bordeaux (France) <https://hal.science/hal-05111581>.
3. **Performance evaluation of adaptive introgression classification methods.** J. Romieu, G. Camarata, P.-A. Crochet, *M.d. Navascués*, R. Leblois, F. Rousset (2024) in *Mathematical and Computational Evolutionary Biology*, MCEB, Saint-Martin-de-Londres (France) <https://hal.inrae.fr/hal-04713611>.
4. **SelNeTime: A new method inferring demography and selection from genomic time series data.** M. Uhl, S. Boitard, *M.d. Navascués*, B. Servin (2024) in *conservgenomics: Conservation Genomics Paris*, Muséum National d'Histoire Naturelle, Paris (France) <https://hal.inrae.fr/hal-04659055>.
5. **Joint inference of demography and selection from temporal population genomic data via approximate Bayesian computation.** V.A.C. Pavinato, S. De Mita, J.-M. Marin, *M.d. Navascués* (2021) in *Probabilistic Modeling in Genomics (Virtual)*, Cold Spring Harbor Laboratory, Online (USA) [doi:10.6084/m9.figshare.23943312](https://doi.org/10.6084/m9.figshare.23943312);
6. **Joint inference of effective population size and genetic load from temporal population genomic data.** V.A.C. Pavinato, S. De Mita, J.-M. Marin, *M.d. Navascués* (2021) in *Ancient Biomolecules of Plants, Animals, and Microbes*, Wellcome Connecting Science, Online (UK) [doi:10.6084/m9.figshare.23943828](https://doi.org/10.6084/m9.figshare.23943828);
7. **Detection of loci under selection from temporal population genomic data through ABC random forest.** V.A.C. Pavinato, J.-M. Marin, *M.d. Navascués* (2018) in *Journées Scientifiques du LabEx Numev*, LabEx Numev, Montpellier (France) <https://hal.inrae.fr/hal-02787307>.
8. **Genetic diversity in selfing populations: a case study in *Medicago truncatula*.** M. Jullien, J. Ronfort, *M.d. Navascués*, L. Gay (2018) in *II Joint Congress on Evolutionary Biology*, European Society for Evolutionary Biology, American Society of Naturalists, Society for the Study of Evolution, Society of Systematic Biologists, Montpellier (France) [doi:10.6084/m9.figshare.13022732](https://doi.org/10.6084/m9.figshare.13022732).
9. **Temporal FST genome scans: the case of partially selfing populations.** *M.d. Navascués*, A. Becheler, L. Gay, J. Ronfort, R. Vitalis (2018) in *II Joint Congress on Evolutionary Biology*, European Society for Evolutionary Biology, American Society of Naturalists, Society for the Study of Evolution, Society of Systematic Biologists, Montpellier (France) [doi:10.6084/m9.figshare.13010312.v1](https://doi.org/10.6084/m9.figshare.13010312.v1);
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12. **Tracking adaptation in selfing plant populations.** M.d. Navascués, A. Becheler, M. Jullien, L. Frachon, C. Libourel, R. Villoutreix, J. Ronfort, F. Roux, R. Vitalis, L. Gay (2016) in *Evolutionary genomics and systems biology: bringing together theoretical and experimental approaches*, Jacques Monod Conference, Roscoff (France) [doi:10.6084/m9.figshare.4036098](https://doi.org/10.6084/m9.figshare.4036098);
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15. **Seed collections to study plant evolution.** L. Gay, J. Dhinaut, J.-M. Prosperi, M.d. Navascués, J. Ronfort (2015) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMBE, Vienna (Austria) <https://hal.inrae.fr/hal-02795320>.
16. **Demographic inference using skyline plots on approximate Bayesian computation.** M.d. Navascués, C. Burgarella (2012) in *Annual Meeting of the Society for Molecular Biology et Evolution*, SMBE, Dublin (Ireland) [doi:10.6084/m9.figshare.1108025](https://doi.org/10.6084/m9.figshare.1108025).
17. **Inférence démographique utilisant des “skyline plots” avec le calcul Bayésien approché.** M.d. Navascués (2012) in *Le Petit Pois Déridé*, Groupe d’Etude de Biologie et Génétique des Populations, Avignon (France) <https://hal.inrae.fr/hal-02745827>.
18. **Demographic inference using skyline plots on approximate Bayesian computation.** M.d. Navascués, C. Burgarella (2012) in *Mathematical and Computational Evolutionary Biology*, MCEB, Saint-Martin-de-Londres (France).
19. **Étude d’un système de reproduction combinant auto-incompatibilité gamétophytique et asexualité.** M.d. Navascués, S. Stoeckel, S. Mariette (2010) in *Écologie 2010*, Société Française d’Écologie, Montpellier (France) [doi:10.6084/m9.figshare.1108015](https://doi.org/10.6084/m9.figshare.1108015).
20. **Mutation rate estimates for 110 Y-chromosome STRs combining population and father-son pair data.** M.d. Navascués, C. Burgarella (2010) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMBE, Lyon (France) [doi:10.6084/m9.figshare.1108022](https://doi.org/10.6084/m9.figshare.1108022).
21. **Distinguishing recent introgression from ancestral polymorphism as the origin of shared polymorphisms and application on the *Drosophila simulans* species complex.** M.d. Navascués, D. Legrand, C. Campagne, M.-L. Cariou, F. Depaulis (2009) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMBE, Iowa City (USA) [doi:10.6084/m9.figshare.1108024](https://doi.org/10.6084/m9.figshare.1108024).
22. **Discrimination between ancestral polymorphism and recent introgression as causes of shared polymorphic sites between species.** M.d. Navascués, F. Depaulis (2008) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMBE, Barcelona (Spain) [doi:10.6084/m9.figshare.1108019](https://doi.org/10.6084/m9.figshare.1108019).

23. **Demographic expansions in conifer and human populations characterized by linked microsatellite data.** C. Burgarella, M.d. Navascués (2008) in *Annual Meeting of the Society for Molecular Biology and Evolution*, SMOBE, Barcelona (Spain) [doi:10.6084/m9.figshare.1108021](https://doi.org/10.6084/m9.figshare.1108021).
24. **Evolution of inbreeding depression in species combining self-incompatibility and partial asexual reproduction.** M.d. Navascués, S. Stoeckel, S. Mariette (2007) in *Congress of the European Society of Evolutionary Biology*, ESEB, Uppsala (Sweden) [doi:10.6084/m9.figshare.1108016](https://doi.org/10.6084/m9.figshare.1108016).
25. **Characterization of historical demographic expansions from linked microsatellite data.** M.d. Navascués, O. Hardy (2007) in *Evolutionary Genomics*, Jacques Monod Conference, Roscoff (France) [doi:10.6084/m9.figshare.1108020](https://doi.org/10.6084/m9.figshare.1108020).
26. **Evaluation of the power of assignment methods on chloroplast microsatellites for provenance identification of reforested stands.** M.d. Navascués (2006) in *Population Genetics and Genomics of Forest Trees: from Gene Function to Evolutionary Dynamics and Conservation*, IUFRO, Alcalá de Henares (Spain) [doi:10.6084/m9.figshare.1108018](https://doi.org/10.6084/m9.figshare.1108018).
27. **Diversidad y flujo genético en plantaciones de Pinus canariensis.** M.d. Navascués, B.C. Emerson (2006) in *IV Jornadas Forestales de la Macaronesia*, Gobierno de Canarias, Governo dos Açores, Arena, Breña Baja, La Palma (Spain) [doi:10.6084/m9.figshare.1108023](https://doi.org/10.6084/m9.figshare.1108023).
28. **Chloroplast microsatellite evolution in conifers: measures of genetic diversity and the effects of homoplasy.** M.d. Navascués, B.C. Emerson (2003) in *Congress of the European Society for Evolutionary Biology*, ESEB, Leeds (UK) [doi:10.6084/m9.figshare.1108017](https://doi.org/10.6084/m9.figshare.1108017).
29. **Chloroplast microsatellite evolution in conifers.** M.d. Navascués, B.C. Emerson (2003) in *Postgraduate Poster Competition*, Association of Applied Biologists, London (UK) [doi:10.6084/m9.figshare.1108017](https://doi.org/10.6084/m9.figshare.1108017).

seminars

1. **Analysis of Dated Archaeological Record through Approximate Bayesian Computation** Population Genetics group, UMR CBGP (Montpellier France) 14/11/2022
2. **Using Approximate Bayesian Computation for Demographic Inference from Radiocarbon Data** Human Evolution Program, Uppsala University (Sweden) 20/10/2022
3. **Science Communication using Stop-Motion Videos** PhD student meeting “Journal club” at UMR CBGP (Montpellier, France) 28/09/2022
4. **Using Approximate Bayesian Computation for Demographic Inference from Radiocarbon Data** Human Evolution Program, Uppsala University (Sweden) 19/05/2022
5. **Estimation de l’histoire évolutive récente d’une population à partir de données génomiques temporelles** DevOCGen project meeting (Saint Martin de Londres, France) 12/04/2022
6. **Simulation of quantitative traits evolution: burn-in for mutation-drift-stabilizing selection balance** Population Genetics group, UMR CBGP (Montpellier France) 18/03/2022
7. **How to simulate sequencing and genotyping errors from true genotypes** Mattias Jakobsson Lab, Human Evolution Program, Uppsala University (Sweden) 07/02/2022

8. **Breaking bad habits in scientific computing** Human Evolution Program, Uppsala University (Sweden) 29/04/2021
9. **Population genetic forecasting from temporal samples** Mattias Jakobsson Lab, Human Evolution Program, Uppsala University (Sweden) 22/09/2020
10. **Estimating $\frac{1}{N_e}$: Estimating “a bunch of different numbers that are central to our ideas [...] and call them all the same thing” which “is neither effective nor a population size”** Human Evolution Program, Uppsala University (Sweden) 13/02/2020
11. **Inférence statistique en génétique des populations : données temporelles et régime autogame** Séminaires UMR Biogeco, Cestas (Pierroton, France) 21/06/2019
12. **Joint inference of adaptive and demographic history from temporal population genomic data** Réunion du réseau de génétique du Département EFPA (Montpellier, France) 16/05/2019
1. **Changes in genetic diversity through time in selfing populations** Human Evolution Program, Uppsala University (Sweden) 08/06/2017
2. **Tracking adaptation to environmental changes: selfing and detection of loci under selection** Seminar at UMR DIADE (Montpellier, France) 27/03/2017
3. **Tracking adaptation to environmental changes: selfing and detection of loci under selection** Seminar at BioPASS (Dakar, Senegal) 30/06/2016
4. **SelfAdapt — Adaptation to climate change under selfing: Detecting selection with temporal samples** Seminar for the Metaprogram ACCAF (Paris, France) 10/03/2016
5. **Tracking adaptation to environmental changes in experimental or monitored populations: evaluation of a method to detect loci under selection** Seminar of the UMR CBGP (Montpellier, France) 30/06/2015
6. **Demographic inference from genetic data with approximate Bayesian skyline plots** Seminar at Laboratoire Évolution et diversité biologique (Toulouse, France) 19/11/2014
7. **Limites des genome-scans pour la détection de locus sous sélection chez les espèces autogames** Journées ARCAD (Montpellier, France) 15/10/2014
8. **Généalogies** Seminar of the UMR CBGP (Montpellier, France) 20/11/2012
9. **Demographic Inference using Skyline Plots on ABC** ANR project meeting (Montpellier, France) 03/07/2012
10. **Skyline plots from approximate Bayesian computation analyses: an intuitive and flexible representation of demographic dynamics** Groupe de réflexion Génétique des populations et phylogéographie CBGP (Montpellier, France) 31/05/2011
11. **Skyline plots from approximate Bayesian computation analyses: an intuitive and flexible representation of demographic dynamics** Department of Forest Genetics, CIFOR-INIA (Madrid, Spain) 27/05/2011
12. **Étude d'un système de reproduction combinant auto-incompatibilité gamétophytique et asexualité** Séminaire d'Écologie et Évolution, Centre d'Ecologie Fonctionnelle et Evolutive CNRS UMR5175 (Montpellier, France) 07/05/2010

13. **Trans-specific polymorphism: isolation (incomplete lineage sorting) vs. migration (introgression)** Seminar at the Équipe de Éco-Évolution Mathématique, CNRS UMR7625 (Paris, France) 13/10/2009
14. **Trans-specific polymorphism: isolation (incomplete lineage sorting) vs. migration (introgression)** Seminar at the UMR CBGP, INRA (Montpellier, France) 14/09/2009
15. **Reproductive system evolution in partially clonal, self-incompatible plants** Seminar at Istituto di Genetica Vegetale, CNR (Sesto Fiorentino, Firenze, Italy) 21/04/2009
16. Contribution to presentation **Projet Transbiodiv : Biodiversité trans-spécifique neutre et fonctionnelle : développements théoriques et quantification chez des organismes modèles** Séminaire de suivi du programme ANR Biodiversite (Paris, France) 05/03/2009
17. **Discriminación entre polimorfismo ancestral e introgresión reciente mediante patrones de desequilibrio de ligamiento en test no-paramétricos** Seminario CIFOR-INIA (Madrid, Spain) 18/07/2008
18. **Approximate Bayesian computation** ANR Project Transbiodiv Workshop (Paris, France) 01/06/2008
19. **Distinguishing shared ancestral polymorphism from recent introgression** Séminaire Laboratoire Évolution, Génomes et Spéciation CNRS (Gif-sur-Yvette, France) 18/06/2008
20. **Detection of clustering of shared polymorphisms: introgressed blocks** ANR Transbiodiv Meeting (Paris, France) 25/04/2008
21. **Propagation asexuée et maintien du système d'auto-incompatibilité** Séminaire inter-équipes UMR 7625 (Paris, France) 04/04/2008
22. **Characterization of demographic expansions from linked microsatellite data** TIMC-IMAG Seminar (Grenoble, France) 07/02/2008
23. **Evaluation of substitution rates estimates from BEAST: robustness to violations of the coalescent model** Journées scientifiques RTP Paléogénétique de l'Homme et de son environnement, 19–20 novembre 2007, Paris (France) 19/11/2007
24. **Characterization of demographic expansions from linked microsatellite data** Conférence Universitaire de Suisse Occidentale “Estimating Demographic Parameters from Genetic Data” Workshop (La Foully, Switzerland) 07/06/2007
25. **Characterization of demographic expansions from linked microsatellite data: dealing with homoplasy and the case of Canary Island pine** Séminaire Évolution Moléculaire (Paris, France) 23/04/2007
26. **Characterization of demographic expansions from linked microsatellite data: dealing with homoplasy and the case of Canary Island pine** Seminar at the Équipe de Éco-Évolution Mathématique, CNRS UMR7625 (Paris, France) 13/03/2007
27. **Detection of population expansions with linked microsatellites** EEE-LUBIES Seminar, Université Libre de Bruxelles (Belgium) 19/10/2006
28. **On volcanoes, beetles and pines** BioGeCo Seminar, INRA-Pierroton (Bordeaux, France) 31/07/2006

29. **Genetic introgression between endemic *Pinus sylvestris* var. *nevadensis* and *P. sylvestris* plantations studied with chloroplast microsatellites** At Course on Molecular Marker Analysis of Plant Population Structure and Processes by Royal Veterinary and Agricultural University (Copenhagen, Denmark) 05/2006
30. **Population genetics of the Canary Islands pine** Department of Forest Genetics CIFOR-INIA (Madrid, Spain) 02/2006
31. **Population genetics of the Canary Islands pine** Molecular Ecology laboratory at the University of East Anglia (Norwich, UK) 06/05/2005
32. **Genetic diversity in the Canary Island pine tree** Research Open-Day “Bio-Colloquium 2004” School of Biological Sciences, University of East Anglia (Norwich, UK) 03/06/2004
33. **Measuring genetic diversity in pine forest** Centre for Ecology, Evolution and Conservation at the University of East Anglia (Norwich, UK) 01/04/2003
34. **Simulation methods in biology** Molecular Ecology laboratory at the University of East Anglia (Norwich, UK) 30/01/2004
35. **Homoplasmy in chloroplast microsatellites** Department of Forest Genetics CIFOR-INIA (Madrid, Spain) 14/01/2004
36. **Homoplasmy: why on Earth (and on other distant planets) have intelligent life forms evolved into humanlike organisms?** Molecular Ecology laboratory at the University of East Anglia (Norwich, UK) 04/10/2002
37. **Biogeography of *Pinus canariensis*** Centre for Ecology, Evolution and Conservation at the University of East Anglia (Norwich, UK) 21/05/2002
38. **Genética de poblaciones del pino canario** Departamento de Biología at Universidad Autónoma de Madrid (Spain) 03/04/2002
39. **From the Beatles to the Red Hot Canary Pine-Trees** Molecular Ecology laboratory at the University of East Anglia (Norwich, UK) 16/11/2001

PROFESIONAL ACTIVITIES

Administrative duties

Teaching

teaching materials

1. **Crash Course on Approximate Bayesian Computation in Population Genetics** *M.d. Navascués* (2018–2023) [doi:10.5281/zenodo.1435503](https://doi.org/10.5281/zenodo.1435503)

courses

professional

1. **Crash Course on Approximate Bayesian Computation in Population Genetics** Graduate School of Biology, Uppsala University (Sweden) 05/2021
2. **Estimation of effective population size** GenTree Training Workshop on “Methods in Forest Conservation Genetics”. Madrid (Spain) 2h 12/2019
3. **Calcul bayésien approché (ABC)** Formation collective Cirad-UMR AGAP. Montpellier (France) 10h 10/2018
4. **Approximate Bayesian Computation** Jakobsson lab, Uppsala University. Uppsala (Sweden) 10h 05/2017
5. **Inférence de l’histoire des populations avec la méthode *Approximate Bayesian Computation* (ABC)** Réunion du réseau de génétique du Département EFPA-INRA. Nancy (France) 4h 11/2014
6. **Identificación de la procedencia de material forestal de reproducción mediante el uso de métodos de asignación basados en marcadores moleculares** III Curso Internacional INIA-AECI “Mejora y Conservación de Recursos Genéticos Forestales” (Madrid, Spain) 1.5h 11/2006

university

1. **MU 512 Données, Statistiques et Analyse des données 2** Teaching assistant at laboratory sessions “Statistiques avec R”, Université Pierre et Marie Curie (Paris, France) 02/2009
2. **BIO-1A11 Molecular Biology and Genetics** Teaching assistant at laboratory sessions “Cytogenetics”, University of East Anglia (Norwich, UK) 2004–2005
3. **BIO-1A03 Biodiversity** Teaching assistant at laboratory sessions “Protozoa” and “Invertebrate Locomotion”, University of East Anglia (Norwich, UK) 2003–2004
4. **BIO-1A03 Biodiversity** Teaching assistant at laboratory sessions “Protozoa”, University of East Anglia (Norwich, UK) 2002–2003
5. **BIO-1A04 Evolution, Behaviour and Ecology** Teaching assistant at laboratory sessions “Population Genetics”, University of East Anglia (Norwich, UK) 2001–2002

supervision and mentoring

postdoc

1. **Adaptive introgression in Pearl Millet** Maud Duranton (2022–2024) Postdoc at project Pami-ADAPT. Co-supervisor
2. **Detection of loci under selection from temporal population genomic data through ABC random forest** Victor A.C. Pavinato (2018–2019) Postdoc at project ABCSelection. Main supervisor

doctoral

1. **Estimation of the recent evolutionary history of a population from temporal genomic data** Paul Bunel (2024–) PhD thesis. Funding: INRAE-PEPR Agroécologie et Numérique. INP Toulouse (France) Co-supervisor
2. **Prediction of genomic offset and applications to synthetic populations** Sebastien Geneste (2024–) PhD thesis. Funding: INRAE. École doctorale GAIA, Institute Agro (France) Co-supervisor
3. **Prediction of genomic offset and applications to synthetic populations** Mathilde Delmond (2022–2023) PhD thesis (discontinued). Funding: INRAE. École doctorale GAIA, Institute Agro (France) Co-supervisor
4. **Development and evaluation of methods for the inference of adaptive introgression** Jules Romieu (2020–2025) PhD thesis. Funding: ANR. École doctorale GAIA, Université de Montpellier (France) Co-supervisor
5. **Analyse temporelle de la diversité en régime autogame : Approches théorique et empirique** Margaux Jullien (2015–2019) PhD thesis. Funding: INRA. École doctorale GAIA, Institute Agro (France) Co-supervisor

master

1. **Tri spatial et maintien du polymorphisme dans les populations panmictiques** Assinie Zana (2026) MSc thesis (M2). Formation d'ingénieur en bioinformatique et modélisation - INSA Lyon (France) Co-supervisor.
2. **Integrative Demographic Inference** Kévin Salinas (2026) MSc thesis (M2). Master Biologie Évolutive & Écologie. Université de Montpellier (France) Main supervisor
3. **Integrative Demographic Inference** Marie Le Bailly (2025–2006) Gap year internship. Institute Agro, Montpellier (France) Main supervisor
4. **Integrative Demographic Inference** Thomas Lejeune (2025) MSc thesis (M2). Master Biologie Évolutive & Écologie. Université de Montpellier (France) Main supervisor
5. **Exploration de scénarios évolutifs de colonisation des Îles Kerguelen par la truite commune (*Salmo trutta*)** Laura Vallée (2024) MSc thesis (M2). Master Sciences de l'Eau, Parcours Dynamique des Écosystèmes Aquatiques. Université de Pau et des Pays de l'Adour. Pau (France) Co-supervisor
6. **Performance evaluation of adaptive introgression classification methods** Ghislain Camarata (2023) MSc thesis (M2). Master Biologie Évolutive & Écologie. Université de Montpellier (France) Co-supervisor
7. **Simulating Artificial Recombination for a Deep Convolutional Autoencoder** Fredrik Levin (2021) MSc thesis (M2). Uppsala universitet (Sweden). Subject Reader
8. **Detection of archaic introgression without genomic reference from the source population: an ABC approach** Erifyli-Elonti Giaglara (2019) MSc thesis (M2). Master Biologie Évolutive & Écologie. Université de Montpellier (France) Main supervisor

9. **Évaluation de la performance des statistiques pour la détection de l'introggression adaptative chez le mil (*Pennisetum glaucum*)** Omar Belkasseh (2018) MSc thesis (M1). Master Statistique pour les Sciences de la Vie. Université de Montpellier (France) Main supervisor
10. **Limites des genome-scans pour la détection de locus sous sélection chez les espèces autogames~: une étude par simulation** Arnaud Becheler (2014) MSc thesis (M2). Master Biologie Évolutive & Écologie. Université Montpellier II (France) Main supervisor
11. **Diversification of the olive tree (*Olea europaea*) in the Central Sahara and Macaronesia during the Quaternary** Myriam Goudet (2014) MSc thesis (M2). Université Toulouse III, Paul Sabatier (France) Co-supervisor

Professional Development

training

conferences

1. **Ancient DNA symposium** SciLifeLab Ancient DNA unit, Centre for Palaeogenetics, Swedish Museum of Natural History, Stockholm (Sweden). 2022 November 29
2. **SMBE Everywhere GS4: Using Ancestral Recombination Graphs (ARGs) to Infer Evolutionary Processes** Virtual. 2022 September 30
3. **Les Rencontres Scientifiques Arcad : La Diversité des Plantes Cultivées au Cœur des Trasnitions** Montpellier (France), 2021 September 30–October 1
4. **AEET-SFÉ²** Online Conference for Early Career Scientists. 2021 June 9–11
5. **Cycling backward: The G-BiKE workshop on ancient and historical DNA** COST Action G-BiKE. Virtual. 2020 December 17
6. **Unlocking the past: ancient DNA symposium** Uppsala (Sweden). 2019 September 24–25
7. **3^{ème} Colloque de Génomique Environnementale** Montpellier (France). 2015 October 26–28
8. **Phylogenetic approaches to diversification. Inferring evolutionary past from phylogenetic tree shapes.** Center for interdisciplinary research in biology (CIRB) — Collège de France, Paris (France). 2012 October 22-23
9. Annual meeting GDR 1928 **Génomique des populations et génomique évolutive** Toulouse (France). 2010 November 4–5
10. Réunion Annuelle **R-Syst** (INRA), Thonon-les-Bains (France). 2010 November
11. **JOBIM 2010** Journée satellite Biodiversité et Bioinformatique, Montpellier (France). 2010 September
12. **PhyloCom. Un colloque sur la Phylogénie des Communautés** Montpellier (France). 2010 May
13. Annual meeting GDR 1928 **Génomique des populations et génomique évolutive**, Institut Pasteur, Paris (France). 2009 Octobre 27–28

14. **ABC in Paris** Université Paris Dauphine, Paris (France). 2009 June 26
15. Royal Society Discussion Meeting **Statistical and Computational Challenges in Molecular Phylogenetics and Evolution**, London (UK). 2009 April
16. Annual meeting GDR 1928 **Génomique des populations et génomique évolutive**, Université Lyon 1, Lyon (France). 2007 December
17. Meeting **ANR Modèles Aléatoires de l'Evolution et du Vivant**, Université de Provence, Marseille (France). 2007 October
18. Meeting **Bioinformatique, Modélisation des Systèmes Biologiques**, Institut Henri Poincaré, Paris (France). 2007 October
19. **3rd International Conference on DNA Polymorphisms in Human Populations “DNA Sampling - Strategies and Design”**, Musée de l’Homme, Paris (France). 2007 March
20. **International Symposium: Wild Flora *Ex Situ* Conservation under the Framework of the Convention on Biological Diversity** Jardín Botánico de Córdoba (Spain). 2005 April

EDUCATION